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Toilet shield

Abstract

A toilet shield comprises a gutter shaped portion with an interior channel, a first arm, and a second arm. The interior channel comprises: a rear vertical wall; a first vertical side wall; and a second vertical side wall. The first arm is coupled to a top edge of the first vertical side wall of the interior channel of the gutter shaped portion. The second arm is coupled to a top edge of the second vertical side wall of the interior channel of the gutter shaped portion, wherein the gutter shaped portion is configured such that the interior channel of the gutter shaped portion opens in response to the first arm and the second arm being extended in generally opposite directions.

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Background/Summary

CROSS-REFERENCE TO RELATED U.S. APPLICATIONS—CONTINUATION (1) This application is a continuation application of and claims priority to and benefit of U.S. patent application Ser. No. 18/147,813, now issued U.S. Pat. No. 11,937,744, filed on Dec. 29, 2022, entitled “Toilet Shield” by James Kramer, assigned to the assignee of the present application, the disclosure of which is hereby incorporated herein by reference in its entirety. (2) U.S. patent application Ser. No. 18/147,813 is a continuation application of and claims priority to and benefit of then U.S. patent application Ser. No. 17/335,190, now issued U.S. Pat. No. 11,596,280, filed on Jun. 1, 2021, entitled “Toilet Shield” by James Kramer, and assigned to the assignee of the present application, the disclosure of which is hereby incorporated herein by reference in its entirety. (3) U.S. patent application Ser. No. 17/335,190 claimed priority to and benefit of the then provisional patent application, Ser. No. 63/033,326, entitled “Toilet Shield,” by James Kramer, with filing date Jun. 2, 2020, which was incorporated by reference therein in its entirety.

BACKGROUND

(1) Hundreds of thousands of people experience diarrhea/loose stools every day in the United States alone. Some have a temporary problem that only happens occasionally. Some have life-long, continual, daily afflictions which may be caused by a condition or disease such as irritable bowel syndrome, Crohn's disease, or malabsorption, to name but a few.

(2) One of the negative consequences associated with this is that it creates an undesirable toilet cleaning problem due to splatter of the loose stool in regions of a toilet that are not easily washed away when the toilet is flushed. For example, liquified feces of the loose stool may spray with force in multiple directions and may also splash/rebound upward from the water in the bowl or from the bowl itself to foul the undersurface of the porcelain toilet rim, the underside of the toilet seat, and can end up traveling even higher on the toilet and in hard-to-reach places. In such instances, much of the sprayed and splattered feces remains in these locations after flushing the toilet.

(3) People thusly afflicted can be embarrassed while at work or in another person's home by having to clean up after themselves which takes time and may be hard to accomplish. Moreover, much of this spraying/splattering of feces occurs in care situations where the person causing it is not the person who is involved with the cleaning of it afterward.

(4) People who have diarrhea/loose stool issues often use a toilet more than once a day. This means that these disgusting cleaning burdens may occur many hundreds of thousands or millions of times a day in the United States alone.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The accompanying drawings, which are incorporated in and form a part of the Description of Embodiments, illustrate various embodiments of the subject matter and, together with the Description of Embodiments, serve to explain principles of the subject matter discussed below.

(2) FIG. **1**A illustrates an example pattern for folding a toilet shield cut from a sheet of flushable paper or other water soluble/dissolvable and flushable material, in accordance with various embodiments.

(3) FIG. **1**B is a front side view of the toilet shield with the arms extended from opposite sides of the top opening of the tube, in accordance with various embodiments.

(4) FIG. **1**C is a top view which illustrates the open lumen (interior) of the tube which occurs in response to extension of the arms in opposite directions from one another, in accordance with various embodiments.

(5) FIG. **1**D is a right side view of the toilet shield with the right side arm extended, in accordance with various embodiments.

(6) FIG. **1**E is a right side view of the toilet shield with the right side arm extended and also bent at an angle, in accordance with various embodiments.

(7) FIG. **1**F is an upper rear perspective view of the toilet shield with the arms extended to open the lumen of the tube, in accordance with various embodiments.

(8) FIG. **2**A illustrates a second example pattern for folding a toilet shield cut from a sheet of flushable paper or other water soluble/dissolvable and flushable material, in accordance with various embodiments.

(9) FIG. **2**B illustrates a front view of the gutter shaped toilet shield of FIG. **2**A after folding on the fold lines, in accordance with various embodiments.

(10) FIG. **2**C is a top view of the gutter shaped toilet shield of FIG. **2**B, in accordance with various

embodiments.
(11) FIG. **2**D is a right side view of the gutter shaped toilet shield of FIG. **2**B, in accordance with various embodiments.
(12) FIG. **2**E is an upper right perspective view of the gutter shaped toilet shield of FIG. **2**B, in accordance with various embodiments.
(13) FIG. **3**A shows a front view of a bracket and hook, in accordance with various embodiments.
(14) FIG. **3**B shows a right side view of a bracket and hook, in accordance with various embodiments.
(15) FIG. **3**C top view of a bracket and hook, in accordance with various embodiments.
(16) FIG. **3**D shows a right side perspective view of a bracket and hook, in accordance with various embodiments.
(17) FIG. **4** shows a front left perspective detail view of the bracket installed on the rim of a toilet bowl with the hook extending outward and down away from the toilet bowl, in accordance with various embodiments.
(18) FIG. **5** shows a front right perspective view of an example of the toilet shield being lowered into the bowl of a toilet, in accordance with various embodiments.
(19) FIG. **6** shows a front left upper perspective detail view of an example of the toilet shield lowered into the bowl of the toilet with the arms extended but not yet secured, in accordance with various embodiments.
(20) FIG. **7**A shows a top view an example of the toilet shield lowered into the bowl of the toilet with the arms extended and bent to secure to hooks of brackets installed on the rim of the toilet bowl, in accordance with various embodiments.
(21) FIG. **7**B shows a top view an example of the toilet shield lowered into the bowl of the toilet with the arms extended and bent to secure to hooks of brackets installed on the rim of the toilet bowl, in accordance with various embodiments.
(22) Unless specifically noted, the drawings referred to in this Brief Description of Drawings should be understood as not being drawn to scale. Herein, like items are labeled with like item numbers.

DESCRIPTION OF EMBODIMENTS

(23) Reference will now be made in detail to various embodiments of the subject matter, examples of which are illustrated in the accompanying drawings. While various embodiments are discussed herein, it will be understood that they are not intended to limit to these embodiments. On the contrary, the presented embodiments are intended to cover alternatives, modifications and equivalents, which may be included within the spirit and scope the various embodiments as defined by the appended claims. Furthermore, in this Description of Embodiments, numerous specific details are set forth in order to provide a thorough understanding of embodiments of the present subject matter. However, embodiments may be practiced without these specific details. In other instances, well known methods, procedures, components, and circuits have not been described in detail as not to unnecessarily obscure aspects of the described embodiments.

Overview of Discussion

(24) As previously described, people who have chronic diarrhea/loose stools use a toilet more than once a day and create a cleaning problem due to un-flushable fecal splatter in the toilet bowl and other regions of the toilet.
(25) Toilet Shields are not designed to eliminate all immediate cleaning but rather to decrease it substantially and to limit cleaning to areas that are easily reached with a toilet brush requiring little effort. The toilet shield described herein serves to solve or minimize this cleaning problem by reducing and/or preventing splatter of feces. Toilet shields described herein are designed to accomplish protection of a toilet from becoming fouled by liquid and/or splattering feces while using an economy of flushable paper or other flushable material to avoid becoming a burden on septic systems and waste treatment plants. For example, in a flushable paper embodiment, each

single use toilet shield is made of an amount of flushable paper which may be equal to as little as several sheets of toilet paper.

(26) The extensive amount of cleaning required in these situations where a toilet is fouled with fecal splatter when there is no toilet shield requires much more paper to be used (than would be contained in a toilet shield made of flushable paper). Additionally, the cleaning of a toilet fouled in the absence of a toilet shield may also result in the use of paper (e.g., paper towels) that is not designed to be flushable, thus creating a burden on septic systems and waste treatment plants if it is flushed.

(27) Herein some example toilet shields, example toilet shielding systems, and example methods of use are described. Generally, the toilet shields supply one or more substantially vertical surfaces suspended within the bowl of a toilet such that the substantially vertical surfaces intercept the stream/splatter of diarrheal feces before they impact with the hard surface of the toilet bowl or the hard surface of the water in the bowl. This interception prevents or reduces the amount of splatter off of the bowl or the water. In situations where a stream of diarrheal feces is predominately directed downward and/or downward/reward toward the rear of a toilet bowl, the substantially vertical surfaces of the toilet shield intercept the stream and reduce/eliminate splatter. This reduces splatter near the rear of the toilet bowl, under the rim of the toilet bowl, and splatter that lands above where the flushing water can reach it such as on the inside rim and seat of the toilet bowl.

Example Toilet Shields

(28) FIG. **1**A illustrates an example pattern for folding a toilet shield **100** which has been cut from a sheet a sheet of flushable paper or other water soluble/dissolvable and flushable material, in accordance with various embodiments. By “flushable” what is meant is that the material will either dissolve, degrade in water, or else not clog or disrupt toilet lines, sewer lines, a sewer system, or a septic system when flushed. Dashed lines **113**, **114**, and **115** in FIG. **1**A illustrate regions where folding occurs to join portion **119**A and **119**C together such that portions **119**A and **119**B form a front interior wall **119** of a tube **110**. For example, portion **119**C may be affixed to portion **119**A by any suitable means to include, but not limited to: gluing, knurling, crimping, and folding. The tube **110** may be expanded to open an interior lumen **111** of the tube **110**. The lumen **111** has a rear interior wall **116**, a left side interior wall **117**, a right side interior wall **118**, and a front interior wall **119** (formed by portions **119**A and **119**B when joined). In some embodiments, the tube **110** may include one or more pleats **112**. Dashed lines in regions **112**A and **112**B in FIG. **1**A illustrate regions where folding may occurs to create expandable pleats **112**A and **112**B in tube **110**. The tube **110** includes at least two arms **120** (**120**A and **120**B) disposed on edges of a top opening of the tube **110**. By folding on dashed fold lines **122** and **123**, arms **120**A and **120**B respectively are caused to extend from the tube **110** at substantially right angles to the vertical length of the tube **110**. In some embodiments, an arm **120** may include one or more adjustment holes **121** (e.g., adjustment holes **121**A & **121**B) for interfacing with a hook of a bracket disposed on the rim of a toilet bowl. In some embodiments, the arms **120** may include an adhesive on a side that is intended to couple with the rim of a toilet bowl. The tube **110** of toilet shield **100** is open on at least one end (e.g., the top, near the arms **120**) for receiving human feces from defecation; and may be open on both ends. Although a cutting/folding pattern is illustrated in FIG. **1**, other mechanisms for manufacturing and/or assembly of the toilet shield **100**, may be employed.

(29) FIG. **1**B is a front side view of the toilet shield **100** with the arms **120** (**120**A and **120**B) extended from opposite sides of the top opening of the tube **110**, in accordance with various embodiments. Extending the arms **120** in generally opposite directions causes the lumen **111** of tube **110** to open. In FIG. **1**B, the interior lumen **111** is visible towards the rear due to this embodiment exhibiting a raised collar on the rear side of the tube **110**.

(30) Further, in some embodiments, the design of the toilet shield **100** allows for a raised collar **150** to rise above the horizontal arms **120** in the rear of the toilet shield only. In some instances, liquid feces under pressure of expulsion from the human body tends to spray toward the rear of the toilet

more than toward the front. The raised collar **150** improves the interface between the defecating person and the shield **100** in order to catch/deflect more liquid feces in such situations.
(31) In some embodiments, on either side of the raised rear collar **150** there is a vertical pleat **112** (**112**A, **112**B). The purpose of this pleat **112** is to discourage tearing of the arm **120** at these attachment sites, to allow for better raising of the collar **150**, to increase the extension of the substantially vertical portion of the toilet shield **100** without adding much more paper/flushable material, and to improve the interface between the defecating person and the toilet shield **100** by decreasing the distance between the two.
(32) FIG. **1**C is a top view which illustrates the open lumen **111** of the tube **110** which occurs in response to extension of the arms **120** (**120**A & **120**B) in generally opposite directions from one another, in accordance with various embodiments. In various tubular embodiments, the tube **110** of a toilet shield **100** (and its interior lumen **111**, when open) may have a generally cylindrical, hexagonal, rectangular, conical, bell, or other tubular shape. The adjustment holes **121** are depicted as circular, but may be formed in other shapes such as oval or rectangular.
(33) FIG. **1**D is a right side view of the toilet shield **100** with the right side arm **120**B extended, in accordance with various embodiments.
(34) FIG. **1**E is a right side view of the toilet shield **100** with the right side arm **120**B extended and also bent at an angle (e.g., approximately a right angle) as it would be when coupled with a hook or other fastening means on the rim of a toilet bowl, in accordance with various embodiments.
(35) FIG. **1**F is an upper rear perspective view of the toilet shield **100** with the arms **120** (**120**A & **120**B) extended to open the lumen **111** of the tube **110**, in accordance with various embodiments.
(36) Toilet shields **100** may be provided in a dispenser which holds multiple toilet shields **100**, may be provided in packages with several toilet shields **100**, or may be provided in packages with a single toilet shield **100**.
(37) FIG. **2**A illustrates a second example pattern for folding a toilet shield **200** which has been cut from a sheet a sheet of flushable paper or other water soluble/dissolvable and flushable material, in accordance with various embodiments. By “flushable” what is meant is that the material will either dissolve, degrade in water, or else not clog or disrupt toilet lines, sewer lines, a sewer system, or a septic system when flushed. By folding down and outward on first arm fold line **222** and second arm fold line **223**, arms **220**A and **220**B are formed. Dashed lines **212**A and **212**B in FIG. **2**A illustrate regions where the pattern may be folded. By folding inward on first vertical fold line **212**A and second vertical fold line **212**B a vertical gutter shape is formed which comprises an interior channel **211**. Absent folding on first vertical fold line **212**A and the second vertical fold line **212**B, the vertical shape is more like a curtain (which may have some curvature or appear to have three sides, depending on the positioning of arms **220**A and **220**B when in use), rather than like a gutter with an internal channel. Channel **211** may be expanded to open when the arms **220**A and **22**B are spread in generally opposing directions. The channel **211** has a rear vertical wall **216**, a left side vertical wall **217**, and a right side vertical wall **218**. It is open on a fourth vertical side and does not create a fully enclosed lumen **111** like toilet shield **100** of FIG. **1**B. The toilet shield **200** of FIG. **2**A includes at least two arms **220** (**220**A & **220**B) disposed on edges of an upper opening of the channel **211**. In some embodiments, an arm **220** (**220** and/or **220**B) may include one or more adjustment holes **221** (e.g., holes **221**A in arm **220**A and/or holes **221**B in arm **220**B) for interfacing with a hook of a bracket disposed on the rim of a toilet bowl. The adjustment holes **221** are depicted as circular, but may be formed in other shapes such as oval or rectangular. In some embodiments, adhesive or other fastening means may be utilized to couple the arms **220** to the toilet. For example an adhesive with a peel off covering may be disposed on one or more surfaces of arms **220**. The channel **211** is open on at least one end (the upper portion near the arms) for receiving human feces from defecation; and may be open on both ends. Although a cutting/folding pattern is illustrated in FIG. **2**A, other mechanisms for manufacturing and/or assembly of the toilet shield **200**, may be employed.

(38) FIG. **2**B illustrates a front view of the gutter **210** shaped toilet shield **200** of FIG. **2**A after folding on the fold lines **212**A, **212**B, **222**, and **223**, in accordance with various embodiments. Both the exterior gutter **210** shape and its interior channel **211** are visible. The arms **220** (**220**A & **220**B) may be fastened to brackets or otherwise secured/coupled to a toilet seat or rim of a toilet bowl in order to suspend the toilet shield **200** in place within the bowl of a toilet in a similar manner to that which has been previously described and which is depicted in FIGS. **3**-**6**B. In some embodiments, the bottom edges of the rear vertical wall **216**, the left side vertical wall **217**, and the right side vertical wall **218** trail into the water in the toilet bowl. Lower portions of the substantially vertical walls (**216**, **217**, and **218**) which trail into the water of the toilet bowl pull downward on the toilet shield **200** help to hold the toilet shield **200** in place and give the toilet shield **200** some additional structural integrity. The rear wall **216**, side walls **217** and **218**, and any lower portions of the toilet shield **200** floating or suspended in the water of the toilet bowl serve to decelerate falling/streaming fecal material from a bowel movement, which results in in less upward splashing and splattering of fecal material. This greatly reduces fecal splatter and thus the reduces embarrassment and/or reduces the cleaning burden associated with excessive fecal splatter in a toilet bowl.
(39) Further, in some embodiments, the design of the toilet shield **200** allows for a raised collar (not depicted, but similar to raised collar **150** of FIG. **1**B) to rise above the horizontal arms **220** in the rear of the toilet shield only. In some instances, liquid feces under pressure of expulsion from the human body tends to spray toward the rear of the toilet more than toward the front. The inclusion of a raised collar may improve the interface between the defecating person and the shield **200** in order to catch/deflect more liquid feces in such situations.
(40) FIG. **2**C is a top view of the gutter shaped toilet shield **200** of FIG. **2**B, in accordance with various embodiments.
(41) FIG. **2**D is a right side view of the gutter shaped toilet shield **200** of FIG. **2**B, in accordance with various embodiments.
(42) FIG. **2**E is an upper right perspective view of the gutter shaped toilet shield **200** of FIG. **2**B, in accordance with various embodiments.
(43) Toilet shields **200** may be provided in a dispenser which holds multiple toilet shields **100**, may be provided in packages with several toilet shields **200**, or may be provided in packages with a single toilet shield **200**.
(44) As previously discussed, toilet shields **100** and **200** are made of flushable paper or other water soluble/dissolvable and flushable material such as PVA (polyvinyl alcohol) film. By “flushable” what is meant is that material from which the toilet shield **100**/**200** is formed is safe to flush down a toilet in a similar fashion to toilet paper, without risk of clogging plumbing or causing harm to a wastewater treatment plant. Such material which is safe to flush in this manner is referred to as “flushable material.” The toilet shield **100**/**200** comprises a vertical tube **110**, gutter **210**, or curtain, which is open on one or both ends, and arms **120**/**220** on disposed on the edges of an open end. One or more of tube **110**, gutter **210**, or curtain and the arms **120**/**220** may be composed of flushable material, in some embodiments. The tube **110**, gutter **210**, or curtain may be composed of a different flushable material than the arms **120**/**220**, in some embodiments. In some embodiments, certain portions such as seams, pleats **112**, fold lines, adjustment holes **121**/**221** or potential weak spots may be coated with a biodegradable and/or flushable overspray. The overspray may add structural integrity to fail points. The overspray may provide other advantages such providing a pleasant scent and/or adding a measured amount of water resistance (i.e., slowing the dissolvability) of a portion of the toilet shield **100**/**200**.
(45) In various embodiments, the substantially vertical sides of the toilet shield **100**/**200** may be smooth or faceted. The end of the toilet shield which is configured with arms **120**/**220** is the top end of the toilet shield **100**/**200** when in use. The toilet shield **100**/**200** may be stored in a flattened state prior to use, and the lumen **111**, channel **211**, or curtain opened/deployed for use by extending the arms **120**/**220** away from each other in directions which are generally orthogonal to the

direction in which the toilet shield **100**/**200** extends into the bowl of the toilet. In some embodiments the toilet shield **100**/**200** may be folded and packaged so that it may be easily transported by a person, such as in a purse, wallet, or pocket. In other embodiments, a plurality of toilet shields **100**/**200** may be packaged together.

(46) In some embodiments, the design of the toilet shield **100**/**200** with regard to the attachment of the horizontal arms **120**/**220** to the substantially vertical portion (e.g., the tube **110** or gutter **210**) allows for the lumen **111**, channel **211**, or curtain of the toilet shield **100**/**200** to open/deploy when the arms **120**/**220** are pulled away from each other in generally opposite directions.

Example Bracket and Hook for Securing Arms of a Toilet Shield

(47) FIG. **3**A shows a front view of a bracket **300** and hook **310**, in accordance with various embodiments. The bracket **300** may be installed on the rim of a toilet bowl. The hook **310**, which is coupled with the bracket **300** is configured to engage with an arm **120**/**220** of a toilet shield, either by piercing the material of the arm or by being inserted in a preconfigured adjustment hole **121**/**221** of the arm **120**/**220**. Although hooks **310** are depicted as being bent, thy may be straight in some embodiments. Although hooks **310** are depicted as circular in shape they may be formed in other suitable shapes, such as rectangular, square, oval, etc. One or more ridges or standoffs may be configured onto a surface of the bracket **300** in order to provide an interface between a bracket **300** and a toilet shield lowered on top of a bracket **300**.

(48) These brackets **300** snap into place by virtue of their elastomeric quality. In some embodiments, the brackets include a lip which secures to the underside of the rim of the toilet. On the outward side of the toilet, on the outward facing lateral sides, each bracket **300** has a hook **310** extending outward, away from the toilet with the end of the hook **310** pointing downward so that the hook **310** is open on the bottom. These brackets **300** can be left in place for months or can be placed for one time use and then removed.

(49) FIG. **3**B shows a right side view of a bracket **300** and hook **310**, in accordance with various embodiments.

(50) FIG. **3**C top view of a bracket **300** and hook **310**, in accordance with various embodiments.

(51) FIG. **3**D shows a right side perspective view of a bracket **300** and hook **310**, in accordance with various embodiments.

(52) FIG. **4** shows a front left perspective detail view **400** of a bracket **300-1** installed on the rim **415** of a toilet bowl **410** of a toilet **405** with the hook **310-1** extending outward and down away from the toilet bowl **410**, in accordance with various embodiments.

Example Toilet Shielding System

(53) In some embodiments, a toilet shielding system is comprised of a toilet shield (one embodiment illustrated in FIGS. **1**A-**1**E and another embodiment illustrated in FIGS. **2**A-**2**E) and a bracket **300** which is coupled to a hook **310**. Although toilet shield **100** is depicted in use in FIGS. **5-7**B, use of either of these toilet shield **100** or **200** in such a system would be carried out in a similar fashion to the examples shown in FIGS. **5-7**B.

(54) FIG. **5** shows a front right perspective view of an example of the toilet shield **100** being lowered into the bowl **410** of a toilet **405**, in accordance with various embodiments. Directional arrow **501**, shows the direction in which it is being lowered. Toilet seat **520** is in an “up” position and has not yet been lowered onto rim **415**. Brackets **300-1** and **300-2** are visible installed on the rim **415** of toilet **405**, as is hook **310-1**.

(55) FIG. **6** shows a front left upper perspective view of an example of the toilet shield **100** lowered into the bowl **410** of the toilet **405** (only a partial view of toilet **405** is shown) with the arms **120**A and **120**B extended but not yet secured, in accordance with various embodiments. In some embodiments, the toilet seat **520** may be lowered onto the extended arms **120**A and **120**B to hold them in place and extended, causing the lumen **111** to remain open for receiving human feces during defecation. In some embodiments, an adhesive may be disposed between the arms **120**A and **120**B and the rim **415** of the bowl **410** of the toilet **405** to hold the arms **120**A and **120**B in place

and extended, causing the lumen **111** to remain open for receiving human feces during defecation. A bottom portion of toilet shield **100** may be suspended in or slightly above water **601**. In some embodiments, the extended arms **120**A and **120**B may be secured to the hooks **310** (not visible in FIG. **5**) on either side of the toilet bowl **410**; where the hooks **310** hold the arms **120**A and **120**B in place and extended, causing the lumen **111** to remain open for receiving human feces during defecation and simultaneously shielding the bowl **410** and other portions of the toilet **405** from splatter of loose stool.

(56) FIG. **7**A shows a top view an example of the toilet shield **100** lowered into the bowl **410** of the toilet **405** with the arms **120**A and **120**B extended and bent to secure to hooks **310-1** and **310-2** of brackets **300-1** and **300-2** (not visible in FIG. **7**A) installed on the rim **415** of the toilet bowl **410**, in accordance with various embodiments. Securing the extended arms **120**A and **120**B to the hooks **310-2** and **310-1** on either side of the toilet bowl **410**, as shown, holds the arms **120**A and **120**B in place and causes the lumen **111** to remain open for receiving human feces during defecation. As has been discussed, other means of securing arms **120** may be used instead of hooks **310**. In FIG. **7**A, the toilet seat **520** is still in an upright position.

(57) FIG. **7**B shows a top view an example of the toilet shield **100** lowered into the bowl **410** of the toilet **405** with the arms **120**A and **120**B extended and bent to secure to hooks **310-1** and **310-2** of brackets **300-1** and **300-2** (not visible in FIG. **7**B) installed on the rim **415** of the toilet bowl **410**, in accordance with various embodiments. FIG. **7**B is the same as FIG. **7**A except the toilet seat **520** has been lowered and the toilet **405** is ready for a human to sit on it and defecate into the open lumen **111** of the toilet shield **100**.

Example Method of Use

(58) In some embodiments, first the toilet seat **520** is lifted (if not already lifted) and two plastic brackets **300-1** and **300-2** are attached onto the rim **415** of the toilet bowl **410**, one on either side, across from each other, when looking down at the toilet at **2**:**30** and **9**:**30** on the clockface. An example of this is illustrated in FIG. **5**. The plastic brackets **300** may be left in place and used multiple times. The multiuse plastic brackets **300**, when employed, may be small and made from biodegradable plastic so as not to add to the earth's plastic burden.

(59) To prepare the toilet shield **100**/**200** for use, the opposing arms **120**/**220** are extended outward, with the distal tips of the arms extended generally away from each other. An example of this is shown in FIG. **5**. When the arms **120**/**220** are pulled away from each other the lumen **111**, channel **211**, or curtain of the toilet shield **100**/**200** opens forming an open top end for receiving human feces. In some embodiments, the ends of the arms **120**/**220** are attached to the plastic brackets **300**, such as by piercing the arm **120**/**220** with the hook **310** or by positioning the hook **310** in a preconfigured adjustment hole **121**/**221** in an arm **120**/**220**. An example of this is shown in FIGS. **7**A and **7**B. In other embodiments, the ends of the arms **120**/**220** are attached to generally opposing sides of the toilet bowl **410** and/or the rim **415** of the toilet bowl **410** by other means such as adhesive or hook and loop fasteners.

(60) In some embodiments, attaching the ends of the arms **120**/**220** to the hooks **310** on the plastic brackets **300** causes the toilet shield **100**/**200** to become suspended from the brackets **300** like a hammock between two trees (see for example FIGS. **6**, **7**A, and **7**B). In some embodiments, a portion of the arms **120**/**220**, such as the tips of the arms **120**/**220**, may feature several preconfigured holes **121**/**221** to allow for adjustment of the height of the substantially vertical portion of the toilet shield **100**/**200**. These adjustment holes **121**/**221** also facilitate positioning the toilet shield **100**/**200** within toilet bowls **410** of a variety of widths while ensuring the lumen **111**, channel **211**, or curtain is adequately opened to receive expelled human feces.

(61) The toilet seat **520** is closed/lowered and the toilet shield **100**/**200** is ready for use. An example of this is illustrated in FIG. **7**B. This can be done ahead of time (e.g., an hour in advance) as may occur in care giving situation or immediately prior to use.

(62) In other embodiments, other fastening/suspension means may be employed. For example, in

addition to or in alternative to the use of the bracket **300** and hook **310** on opposite sides of the toilet bowl **410**, an adhesive may be employed between the rim **415** of the toilet bowl **410** and the arms **120**/**220** of the toilet shield **100**/**200**. In some embodiments, for example, an adhesive strip is incorporated on the arms **120**/**220** of the toilet shield **100**/**200** during manufacturing.

(63) In use, the toilet shield **100**/**200** collects liquid/splattered feces that does not drop vertically preventing it from fouling the toilet. The toilet shield **100**/**200** also slows the descent of feces that may be falling or spewing under pressure. When falling feces hits the water **601** some splashes upward and is big source of cleaning problems. The toilet shield **100**/**200** discourages this splashing upward by lessening the amount of feces that hits the water **601** directly and by slowing the descent of feces that does hit the water **601**. The toilet shield also catches feces that are splashing upward from the water **601** impeding and/or preventing the splashed feces from reaching and fouling the toilet.

(64) Once the toilet shield **100**/**200** is used the person using it (or a care giver) can simply release the arms **120**/**220** from the toilet bowl **410**. For example, in some embodiments, this involves releasing the toilet shield **100**/**200** from hooks **310** on the outside of the toilet bowl **410**, one on either side, by lifting the arms **100**/**200** off the hooks **310**. This may accomplished without changing position, with the toilet seat **520** down and without coming in contact with anything untoward.

(65) Once released, the weight of the feces and/or water **601** in the toilet bowl **410** pulls the toilet shield **100**/**200** into the water or the swirling motion of the flushed water **601** pulls it in the toilet bowl **410**, and it is flushed away.

(66) The toilet shield **100**/**200** can also be used without mounting brackets **300** or other attachment means by placing the arms **120**/**220** on top of the toilet seat **520** and holding it in place by the weight of the user. For example, a toilet seat **520** may be lowered onto the rim **415** of a toilet bowl **410**. A toilet shield **100**/**200** is expanded by pulling its arms **120**/**220** in generally opposite directions. The arms **120**/**220** may be draped over the lowered toilet seat **520** at locations which are in the vicinity of 2:30 and 9:30 on the face of a clock (if the rear of the toilet seat is considered 12:00), and after positioning the toilet shield **100**/**200** at the desired elevation within the bowl **410** the excess length of the arms **120**/**220** is then tucked under the toilet seat **520** (e.g., between the toilet seat **520** and the rim **415**). The user then sits on the toilet seat and the arms **120**/**220** which are exposed on the top surface of the seat **520** while defecating. After defecating, the user would exit the toilet **405** at least slightly to release the toilet shield **100**/**200**, and then flush the toilet **405** to flush the toilet shield **100**/**200** in to the sewer/septic system to which the toilet **405** is coupled.

CONCLUSION

(67) The examples set forth herein were presented in order to best explain, to describe particular applications, and to thereby enable those skilled in the art to make and use embodiments of the described examples. However, those skilled in the art will recognize that the foregoing description and examples have been presented for the purposes of illustration and example only. The description as set forth is not intended to be exhaustive or to limit the embodiments to the precise form disclosed. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

(68) In accordance with the embodiments shown, one of ordinary skill in the art will readily recognize that there could be variations to the embodiments and those variations would be within the spirit and scope of what has been disclosed. Accordingly, many modifications may be made by one of ordinary skill in the art without departing from the spirit and scope of the disclosure.

(69) Reference throughout this document to “one embodiment,” “certain embodiments,” “an embodiment,” “various embodiments,” “some embodiments,” or similar term means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of such phrases in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore,

the particular features, structures, or characteristics of any embodiment may be combined in any suitable manner with one or more other features, structures, or characteristics of one or more other embodiments without limitation.

Claims

1. A toilet shield comprising: a gutter shaped portion with an interior channel, the interior channel comprising: a rear vertical wall; a first vertical side wall; and a second vertical side wall; a first arm coupled to a top edge of the first vertical side wall of the interior channel of the gutter shaped portion; and a second arm coupled to a top edge of the second vertical side wall of the interior channel of the gutter shaped portion, wherein the gutter shaped portion is configured such that the interior channel of the gutter shaped portion opens in response to the first arm and the second arm being extended in generally opposite directions.
2. The toilet shield of claim 1, further comprising a raised collar which is disposed between and rises above the first arm and the second arm and is configured to deflect liquid feces expelled from a defecating human toward a rear portion of a toilet.
3. The toilet shield of claim 1, wherein: responsive to the first arm being coupled with a first portion of a rim of a toilet bowl and the second arm being coupled with a second portion of the rim of the toilet bowl, the gutter shaped portion is configured to suspend across and into the toilet bowl with the interior channel of the gutter shaped portion suspended held open to receive feces from human defecation.
4. The toilet shield of claim 3, further comprising: a first adhesive strip disposed on the first arm and configured to adhesively couple the first arm with the first portion of the rim of the toilet bowl; and a second adhesive strip disposed on the second arm and configured to adhesively couple the second arm with the second portion of the rim of the toilet bowl.
5. The toilet shield of claim 3, wherein responsive to the gutter shaped portion being suspended across and into the toilet bowl: bottom edges of the rear vertical wall, the first vertical side wall, and the second vertical side wall are configured to trail into water in the toilet bowl such that a downward pull downward is exerted on the toilet shield to help to hold the toilet shield in place.
6. The toilet shield of claim 1, wherein: the first arm, the second arm, the rear vertical wall, the first vertical side wall, and the second vertical side wall are contiguous portions of a pattern cut from a single sheet of a material.
7. The toilet shield of claim 1, wherein the first arm, the second arm, the rear vertical wall, the first vertical side wall, and the second vertical side wall are formed of a water dissolvable, flushable material.
8. The toilet shield of claim 7, further comprising: a scented flushable overspray applied to the water dissolvable, flushable material.
9. The toilet shield of claim 7, wherein: the water dissolvable, flushable material is paper.
10. The toilet shield of claim 7, wherein: the water dissolvable, flushable material is polyvinyl alcohol film.